

April 2025

Seeking Science VOL 44

# SEEKING SCIENCE

## MONTHLY NEWSLETTER

STEM Action Teen Institution



**Oceanography**  
Audrey Wu

**Engineering Behind  
the MRI**  
Riley Lee

**The Glow of the Sea**  
Denise Lee

# This Edition's Staff

|              |                                          |
|--------------|------------------------------------------|
| Lewis Sun    | <i>Advisor</i>                           |
| Jean Wang    | <i>Advisor</i>                           |
| Owen Chen    | <i>President</i>                         |
| Kenny Wu     | <i>Vice President - Editor</i>           |
| Aidan Hong   | <i>Vice President - Art</i>              |
| Zichun Wang  | <i>Vice President - Public Relations</i> |
| Richard Wang | <i>Quality Control Manager</i>           |
| Brandon Wang | <i>Marketing Manager</i>                 |
| Riley Lee    | <i>Senior Graphics Editor</i>            |
| Kevin Zhang  | <i>Senior Graphics Editor</i>            |
| Denise Lee   | <i>Senior Editor</i>                     |
| Yifan Yang   | <i>Senior Editor</i>                     |
| Mary Liang   | <i>Junior Designer</i>                   |
| Arick Hong   | <i>Junior Designer</i>                   |

|              |                                             |
|--------------|---------------------------------------------|
| Spencer Wang | <i>Junior Designer</i>                      |
| Ethan Liu    | <i>Junior Editor</i>                        |
| Emily Ma     | <i>Junior Editor</i>                        |
| Emily Wang   | <i>Junior Editor</i>                        |
| Audrey Don   | <i>Junior Graphics Editor</i>               |
| Brandon Pian | <i>Junior Graphics Editor</i>               |
| Justin Liu   | <i>Student Outreach</i>                     |
| Audrey Wu    | <i>Student Outreach</i>                     |
| Steve Li     | <i>Student Outreach</i>                     |
| Katherine Hu | <i>Marketing Manager Assistant</i>          |
| Horris Li    | <i>Marketing Manager Assistant</i>          |
| Darren Wang  | <i>Community Reporter Assistant</i>         |
| Jeremy Wu    | <i>Community Reporter Assistant</i>         |
| Howard Pan   | <i>Technology &amp; Resources Help Desk</i> |
| Howard Feng  | <i>Technology &amp; Resources Help Desk</i> |

# Table of Contents

|                            |       |
|----------------------------|-------|
| Future of Digital Twins    | 4-5   |
| Engineering Behind the MRI | 6-7   |
| R Values and U Values      | 8     |
| The Glow of the Sea        | 9-10  |
| Oceanography               | 21-23 |

# Future of Digital Twins

By Richard Wang

---

Digital twins will be revolutionary for industries and will be the reason for the success of many engineering projects in the future. To understand the plethora of benefits and real-life applications that digital twins bring, it is essential to understand what digital twins are. Digital twins are accurate virtual models of physical objects. Take a physical object such as an oil drill. Sensors will be used on important components of the oil drill which provide data about the object's production and performance. Data relayed back to a computer or processing system will be implemented into the virtual copy of the object. With this, simulations can be run, allowing for insight on issues and further evaluation of potential improvements.

There is an interesting history behind digital twins. An early idea of a digital twin shown was when NASA used it in the 1960s for space exploration. However, it would not publicly be introduced and invented in 2002 when Micheal Grieves proposed the idea at a SME (society of manufacturing engineering) conference. The most exciting part of digital twins is how beneficial they can be. It allows for effective research and development. The ample amount of information that is discovered allows for small improvements before further production. Peak efficiency of an item can be reached and digital twins can monitor the system of production. When a product reaches its end of life, the creators can determine which materials can be collected.

Digital twins are used in many applications and areas of industry. They bring benefits to the automotive industry to increase efficiency and vehicle performance. Electricity generation also sees many benefits as digital twins allow for the establishment of timeframes for the maintenance of large equipment such as windmills. Digital twins are present and act like a guide in all stages of the assembling of a product. Civil engineers can use digital twins to work on municipal projects as well.

Digital twins have an unbound future. They continue to collect information with simulations during the development of the product, thus allowing for maximum efficiency. Also, with expanding technology, we may see many changes to digital twins. The future of digital twins seems to be heading on a positive path.

# Engineering Behind the MRI

By Riley Lee

Magnetic Resonance Imaging also known as the MRI, is one of the most important tools used in hospitals today. This allows doctors to see inside your body without the use of surgery or any harmful radiation. What many people don't realize is that an MRI machine is huge example of how different types of engineering work together in order to create fascinating tools or objects. The MRI combines the skills of mechanical, electrical, and materials engineers.



Mechanical engineers design the strong frame and moving parts of the MRI machine. These machines are extremely heavy and very large and they need to be built in a stable and safe position. Mechanical engineers also design the patient bed that moves smoothly in and out of the machine while making sure that

the machine stays very cold. This is because MRI magnets need extremely low temperatures in order to work properly. If an MRI magnet is not at a low temperature, it can lose its superconductivity and experience a “quench” which will cause the entire system related or connected to the MRI to shut down.

Electrical engineers make sure the MRI can send and receive signals. This machine uses strong magnets and radio waves to interact with atoms in your body. Electrical engineers develop the parts that send out these signals and pick up the responses. While they are developing those parts, they need to build the computer systems that turn the signals into images so the doctors can read.

Materials engineers choose the right materials to build the machine. For example, the magnets inside the MRI are made from a unique type of metal that can carry electricity with no resistance, but only when they are super cold. Materials engineers work with these metals and make sure they're safe, effective, and usable. They have to also select or choose other materials that won't interfere with the magnetic field while handling the cold temperatures.

In the end, the MRI machine is more than just medical equipment, it's an amazing result or innovation between different kinds of engineers. Without their teamwork and collaboration, we would never have this safe, reliable, and powerful tool that helps doctors diagnose and treat millions of people every year.

# R Values and U Values

Aidan Hong

---

Typically, in order to maintain heat in a house, engineers will use thermodynamics in order to determine what kind of insulation should be used. While many different types of materials exist to handle this problem, certain materials are better than others. This can be determined through R values and U values

The R value is used to determine a material's ability to resist heat transfer. Typically, a high R value indicates that the material is good at resisting heat transfer, and a low R value indicates it is not very good at resisting heat transfer. When creating insulation, engineers typically aim to maximize the R value. Most of the time, engineers use more than one material to create insulation. In order to determine the total R value, you can simply add up all the R values to get the total R value.

The U value, on the other hand, determines conductivity. This means that a high U value will transfer heat very quickly, and a low U value will not transfer heat very quickly. In terms of insulation, a low u value is desirable as a high u value will transfer heat very quickly. U values and r values are related: to find the u value, you can simply take the reciprocal of the r value.

R values and u values are critical when creating insulation. Especially during summer or winter, these r values and u values are important to engineers: they tell a lot about a material's ability to resist heat, maintain heat, and more. Without these numbers, we will not be able to tell if an insulation works or not through math. These numbers save countless hours that would be otherwise spent on testing and countless dollars from unnecessary testing.

# The Glow of The Sea

By Denise Lee

As you feel the soft sand on your feet as darkness fills the sky, all you can hear are the beautiful, glowing, blue waves splashing on the beach. These stunning waves are called bioluminescent waves. Bioluminescent waves are natural wonders and it is



Licensed under CC0 1.0 Universal  
Phytoplanktons, Gurdin Port, Chabahar, Iran by [Safa.daneshvar](https://creativecommons.org/publicdomain/zero/1.0/deed.en) from Wikimedia Commons  
[https://commons.wikimedia.org/wiki/File:Luminescent\\_beaches\\_in\\_Chabahar4.jpg](https://commons.wikimedia.org/wiki/File:Luminescent_beaches_in_Chabahar4.jpg)

when the ocean sparkles in blue at night. Bioluminescence is nearly impossible to spot in daylight with the naked eye. The WINDY.APP states that “the glow is due to the unicellular *Noctiluca scintillans* algae, which release light in response to external stimuli — in this case to the coastal waves throwing them from side to side.” Bioluminescent waves usually happen during late summer when algae bloom of plankton produces a chemical reaction that includes luciferin and luciferase.

Although the glow of the sea is extremely gorgeous, it can be quite dangerous. According to WebMB, it states that the bioluminescent waves produce toxins that can

sometimes harm marine life, humans, and anything that comes in contact with it. It may also cause skin irritation and burning in the eyes. Swimming or fishing in bioluminescent waves is highly unrecommended. If you eat shrimp, fish, or other marine life that has eaten the algae bloom, you may get food poisoning. Eating a lot of marine life that has been affected by bioluminescence can affect brain functions and memory.

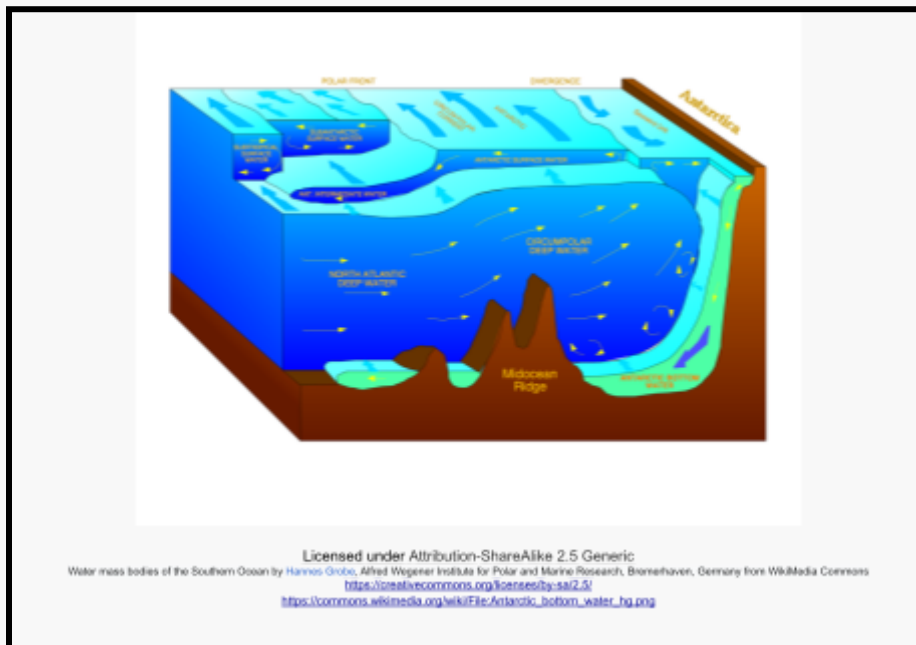
Scientists have been studying bioluminescence to see if it can be used as a light source and an eco-friendly alternative to fossil fuels. Bioluminescence may be another solution to save our planet from climate change or global warming.

Bioluminescent waves are one of the beauties on Earth. All in all, the glow of the sea is as beautiful as the starry nights, but it is as dangerous as it is beautiful.

# Oceanography

By Audrey Wu

Oceanography is the science that studies the ocean, including the water, the animals and plants that live in it, and the land beneath it. It helps us understand how the ocean works and why it is so important to life on Earth. Since the ocean covers more than 70% of our planet, learning about it can teach us a lot about the Earth's climate, weather, and ecosystems. Oceanographers, or scientists who study the ocean, work in many



different areas to explore the deep sea, protect marine life, and solve environmental problems.

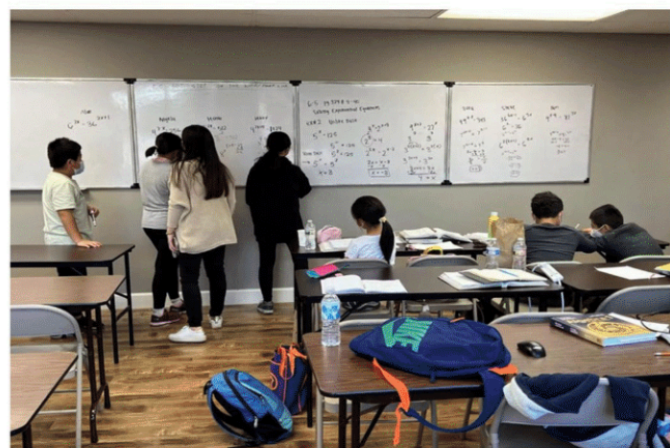
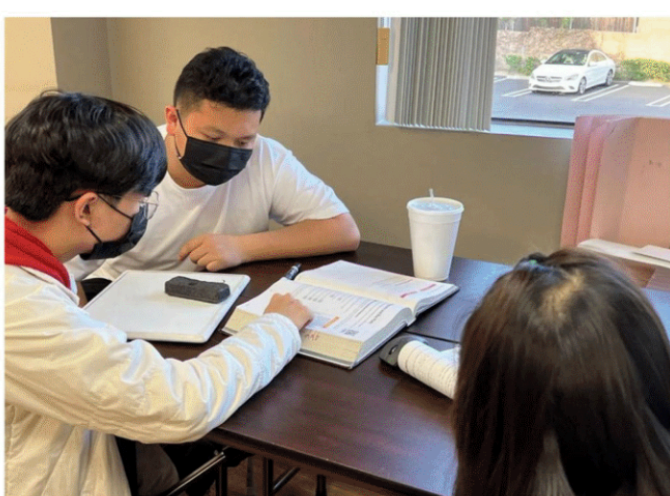
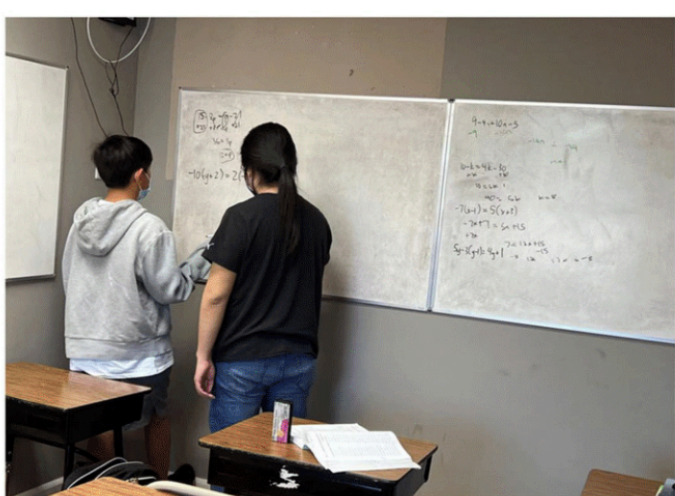
One important part of oceanography is called physical oceanography. This area looks at how

water in the ocean moves, including waves, tides, and currents. These movements are important because they help move heat, nutrients, and oxygen around the world. For example, the Gulf Stream is a powerful ocean current that brings warm water from the tropics up to the east coast of North America and across the Atlantic to Europe. Without it, many places would be much colder. Physical oceanographers also study how the ocean affects weather patterns like hurricanes and El Niño.

Another big part of oceanography includes chemical and biological oceanography. Chemical oceanographers study the salt, gases, and other substances found in ocean water. They also look at how pollution and chemicals like carbon dioxide are changing the ocean. For example, when too much carbon dioxide gets absorbed by the ocean, it becomes more acidic, which can harm sea creatures like coral and shellfish. Biological oceanographers focus on life in the ocean. They study animals like fish, dolphins, and whales, as well as tiny organisms like plankton. These scientists learn how ocean creatures survive, grow, and depend on each other in food chains. Their work helps us protect endangered species and keep ecosystems healthy.

Geological oceanography is the study of the ocean floor. Geological oceanographers explore the underwater landscape, which includes mountains, volcanoes, deep trenches, and valleys. They use special tools and submarines to collect rocks, sediments, and core samples from the seafloor. By studying these materials, they can learn about earthquakes, tsunamis, and how continents move over time. Some geological oceanographers even discover clues about Earth's past climate by looking at layers of mud and fossils deep under





週六下午就是數學大本營時間



科嶺數理電腦學院  
CODING STEM ACADEMY

AI 人工智能教育  
\*最佳推手\*

系統學習 基礎紮實 省時省力 卓越超群

AI人工智能資優兒童班

6-9歲 MIT Scratch, Virtual Robotics

AI人工智能進階班

10-14歲 MIT Inventor, Virtual Robotics

VEX 機器人隊

最有效益的課外活動  
學術競賽與領導才能最大加分

Maker Portfolio

展現實作能力申請一級名校

AP Computer Principle

由編程及網路基礎觀念教起  
全面建立堅實AI能力

AP Computer Science

\* JAVA 程式語言編寫訓練 \*  
邏輯與電腦實務並重

AP Physics 1, 2, C

著重公式練習與演算運用, 同時準備SATII應考

AP Calculus BC, AB

講解清晰海量試題練習 同年應試二科省時省力

數學加強班

Algebra 1, 2 Geometry

物理榮譽班

7-11年級・Honors課程・  
為AP物理作充足準備

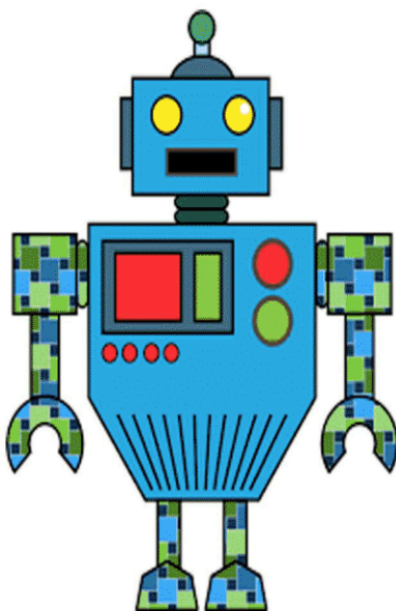
電腦編程基礎班 Java C++ Python

4-12年級為AP Computer 課程準備  
並可參加全國及各項國際AI競賽

SAT 英文寫作班

4-12年級, 閱讀, 文法, 寫作  
\*\* 因材施教 突破盲點 \*\*

教室: 核桃市, 羅蘭崗, 鑽石吧 626-510-0458



2022年賽季將結束，每個小朋友都忙著完成自己的機器人組裝和編程

